6-3 Hexadecimal

Point!

Hexadecimal

(1) ([1]**Hexadecimal**) : A method of representing numbers using the digits 0 to 9 and the alphabet letters A to F. A number represented in hexadecimal is sometimes written with a subscript (16) placed at the bottom right of the number. This is known as a ([2]**hexadecimal number**).

(2) Correspondence Between Decimal, Binary, and Hexadecimal

Decimal	Binary	Hexadecimal
0	0000	0
1	0001	1
2	0010	2
3	0011	3
4	0100	4
5	0101	5
6	0110	6
7	0111	7
8	1000	8

Decimal	Binary	Hexadecimal
9	1001	9
10	1010	([3]**A**)
11	1011	([4]**B**)
12	1100	([5]**C**)
13	1101	([6]**D**)
14	1110	([7]**E**)
15	1111	([8]**F**)
16	10000	([9]**10**)

(3) Conversion between Decimal, Binary, and Hexadecimal

[1] Conversion from binary to hexadecimal

Separate the binary number into groups of 4 digits starting from the least significant bit, convert each group into a hexadecimal value, and arrange these values in sequence.

<Example> Convert the binary number $10011010_{(2)}$ to a hexadecimal number.

Separate into groups of 4 digits starting from the least significant bit: $1001/1010_{(2)}$

$1001_{(2)}$ is $(9_{(16)})$, $1010_{(2)}$ is $(A_{(16)})$, so $10011010_{(2)} = (9A_{(16)})$

[2] Conversion from hexadecimal to binary

Convert each digit of the hexadecimal number into a 4-bit binary number, and arrange those numbers sequentially.

<Example> Express hexadecimal $A4_{(16)}$ in binary.

$A_{(16)}$ is $(1010_{(2)})$, $4_{(16)}$ is $(0100_{(2)})$, so $A4_{(16)} = (10100100_{(2)})$

[3] Conversion from hexadecimal to decimal

Convert the hexadecimal number to a binary number, and then convert that binary number to a decimal number.

<Example> Express the hexadecimal $C6_{(16)}$ in decimal format.

First, to convert the hexadecimal $C6_{(16)}$ to binary, $C_{(16)}$ is $(1100_{(2)})$,

$6_{(16)}$ is $(0110_{(2)})$, so $C6_{(16)} = (11000110_{(2)})$

Next, to convert the binary number $11000110_{(2)}$ into a decimal number,

$11000110_{(2)} = (0 \times 1 + 1 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 0 \times 2^4 + 0 \times 2^5 + 1 \times 2^6 + 1 \times 2^7)$

$= 198$

Warm Up

Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $11011011_{(2)}$ [2] $11110110_{(2)}$ [3] $9E_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $A5_{(16)}$

Explanation

(1) [1] Separate the binary number into groups of four digits starting from the least significant bit, then convert each group into a hexadecimal number and list them sequentially.
$1101_{(2)}$ is $D_{(16)}$, $1011_{(2)}$ is $B_{(16)}$, so $11011011_{(2)} = \underline{DB_{(16)}}$

[2] $1111^{(2)}$ is $F_{(16)}$, $0110_{(2)}$ is $6_{(16)}$, so $11110110_{(2)} = \underline{F6_{(16)}}$

[3] Convert each digit into a 4-bit binary number and list them sequentially.
$9_{(16)}$ is $1001_{(2)}$, $E_{(16)}$ is $1110_{(2)}$, so $9E_{(16)} = \underline{10011110_{(2)}}$

(2) Convert from hexadecimal to binary, and then to decimal.

[1] $A_{(16)}$ is $1010_{(2)}$, $5_{(16)}$ is $0101_{(2)}$, so $A5_{(16)} = 10100101_{(2)}$
Therefore, $10100101_{(2)} = 1 \times 1 + 0 \times 2^1 + 1 \times 2^2 + 0 \times 2^3 + 0 \times 2^4 + 1 \times 2^5 + 0 \times 2^6 + 1 \times 2^7 = \underline{165}$

Try

Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $11010101_{(2)}$ [2] $01110110_{(2)}$ [3] $C5_{(16)}$ [4] $BB_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $31_{(16)}$ [2] $C7_{(16)}$ [3] $9F_{(16)}$ [4] $AB_{(16)}$

Exercise

1 Answer the following questions.

(1) Convert the following binary numbers to hexadecimal, and the hexadecimal numbers to binary.

[1] $01010011_{(2)}$ [2] $01101100_{(2)}$ [3] $10100111_{(2)}$ [4] $11001001_{(2)}$
[5] $2B_{(16)}$ [6] $F1_{(16)}$ [7] $EE_{(16)}$ [8] $DB_{(16)}$

(2) Convert the following hexadecimal numbers to decimal.

[1] $17_{(16)}$ [2] $C2_{(16)}$ [3] $3D_{(16)}$ [4] $AF_{(16)}$